

DATA C104 / History C184D Human Contexts and Ethics of Data

Navigation Tips

- Use the floating Syllabus Navigation menu (above and to the side) to navigate through this syllabus.
 - See the [Course Schedule \(https://bcourses.berkeley.edu/courses/1557692/pages/schedule\)](https://bcourses.berkeley.edu/courses/1557692/pages/schedule) for a weekly outline of lectures and readings.
 - See the [Course Summary](#) below for a list of due dates (use the link at the top right of this page to jump to today's assignment). *All times listed are Pacific Time.*
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Why this course?

Data-driven analytics and artificial intelligence-powered devices now shape innumerable aspects of our lives. Beneath the surface of these technologies, computational and increasingly autonomous techniques that operate on large, ever-evolving datasets are transforming how people act in and know the world. These new analytic tools, algorithmic systems, and computational infrastructures draw from and reconstruct existing societal structures, patterns, and narratives. Sometimes this is obvious, and sometimes it is invisibly so. Data technologies have profound consequences for how we think of ourselves, relate to one another, organize collective life, and envision desirable futures. This course helps you identify and analyze these human stakes, reason about them with others, and shape opportunities to take action toward outcomes that can serve collective well-being.

If you intend to major or minor in Data Science, the course meets the Human Contexts and Ethics (HCE) requirement of Berkeley's Data Science program. It gives you systematic exposure and reflective practice in engaging with the human actions, decisions, and social structures that intrinsically shape your work.

If you don't intend to major or minor in Data Science, the course will jumpstart your knowledge and strengthen your capacity to take part in guiding our datafied world. The course carries the broad, inclusive spirit of Berkeley's Data Science curriculum into the area of human society and collective world-making.

Breadth Requirements

This class has been approved for breadth in Philosophy & Values and Historical Studies and the EECS/LSCS Ethics requirement.

STS Minor

This class counts as an upper-division elective for the [undergraduate minor in Science, Technology, and Society \(https://guide.berkeley.edu/undergraduate/degree-programs/science-technology-society/#minorrequirementstext\)](https://guide.berkeley.edu/undergraduate/degree-programs/science-technology-society/#minorrequirementstext).

Scope and objectives

How do we shape action together in our complex and changing datafied world, aiming at outcomes that improve the human condition? To help you on this path, this course provides an overall introduction to how data science and data technologies -- including data analytics, algorithmic decision systems, machine learning (ML), and artificial intelligence (AI) -- are entangled today with diverse human contexts (histories, institutions, and material bases) and ethics (domains of moral action, collective world-making, and justice).

We will bring historically-grounded perspectives, frameworks from Science, Technology, and Society (STS), and approaches from other disciplines in the humanities and interpretive social sciences to bear on topics that include:

- Doing ethical data science amid shifting definitions of human subjects, consent, and privacy;
- Understanding representation, power shifts, and justice in data-enabled technologies, including predictive analytics, precision (targeted) services, and surveillance technologies;
- Contemporary landscapes of labor and industry; and
- The changing relationship between data, democracy, and public life.

The course aims, first of all, to prepare you to recognize when, where, and how data, analytics, and associated technologies shape and govern the human condition. Further, it aims to provide you with a toolkit with which to think critically about human contexts and ethics issues of data science and data technologies when you encounter them in routine work or daily life, to influence how data science is used to achieve better outcomes for people, and to be able to articulate (for yourself, and to others) what "better" means.

Instructor Information & Communication

Course Instructors

Cathryn Carson | clcarson@berkeley.edu (<mailto:clcarson@berkeley.edu>)

Ari Edmundson | aedmundson@berkeley.edu (<mailto:aedmundson@berkeley.edu>)

Course Coordinator

Maria Soracco | datat104@berkeley.edu (<mailto:datat104@berkeley.edu>)

Curriculum Coordinator

Daniel Kelly | dhkelly@berkeley.edu (<mailto:dhkelly@berkeley.edu>)

All questions about general course logistics (enrollment, general policies, section swaps) should be addressed to the course email:

datat104@berkeley.edu (<mailto:datat104@berkeley.edu>)

Please contact the professors and course coordinator by email. **Please do not use bCourses messages to contact professors or the course coordinator.**

Please address questions about your own learning (discussion section, course assignments, due dates, grading, etc.) initially to your **section instructor**.

If you need to reach out to the professors directly for any reason, please email us together in one email: do not email us the same message separately.

Section Instructor Team

Jose Aguilar | j.aguilar@berkeley.edu (<mailto:j.aguilar@berkeley.edu>)

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Tutors

Anika Edara | anika.edara@berkeley.edu (<mailto:anika.edara@berkeley.edu>)

<mailto:anika.edara@berkeley.edu>) Jehan Jacobs | jehan_jacobs@berkeley.edu (mailto:jehan_jacobs@berkeley.edu)

Ethan Shohet | eshohet@berkeley.edu (<mailto:eshohet@berkeley.edu>)

Tibet Yakut | tyakut@berkeley.edu (<mailto:tyakut@berkeley.edu>)

Office Hours

We encourage you to attend section instructor and professor office hours to discuss course material, grading expectations, and experience in the course, and so we can get to know you better.

The time and location (or Zoom link and passcode) for each instructor's office hours will be listed on the [Discussion Sections and Office Hours](https://bcourses.berkeley.edu/courses/1557692/pages/discussion-sections-and-office-hours) (<https://bcourses.berkeley.edu/courses/1557692/pages/discussion-sections-and-office-hours>) page.

Learning Activities

Orientation

You won't be able to access your course material until you read the Academic Integrity Premise and make your pledge to Academic Integrity—both in the [Orientation \(https://bcourses.berkeley.edu/courses/1557692/pages/orientation-page\)](https://bcourses.berkeley.edu/courses/1557692/pages/orientation-page).

Discussion Sections

Weekly section attendance is mandatory, i.e., required throughout the semester, and is central to successful learning in this course. In discussion you will apply key concepts from lectures to new cases, analyze readings, dig deeper into core issues, and collaborate with others. Section discussions are not just for your own individual learning; you are also contributing to your peers'.

We recognize that there are unexpected contingencies, such as illness, family emergencies, and DSP needs, that may force you to miss your discussion section. You can miss **two** discussion sections without penalty. After that, we provide the option of completing **one** "re-engagement" in place of a missed discussion section to help you make up for missed learning. In this case, re-engagement means attending your GSI's office hours prepared to discuss the readings you missed and how it connects to the lectures. This re-engagement must be completed by the Friday of the following week after your section meets.

Please note that a fourth absence can't be made up and **missing five or more sections may constitute grounds for automatically failing the course**, whether or not you've attended your GSI's office hours to make up for your third absence. If you think you are at risk of exceeding this limit, please be in touch with your section instructor promptly.

In order to facilitate focused discussion and cultivate the skill of attention, **discussion sections in Data C104 are screen-free**. If you think you should be eligible for an exception to this policy, please speak with your GSI.

To enable participation in close reading activities in section have created a [course reader, which you can purchase from Copy Central on Telegraph Avenue](https://copycentralreaders.square.site/product/data-c104-human-contexts-ethics-of-data-edmundson-carson/7DPTZ2W7GZAOFFLJX67UN5V?cp=true&sa=true&sbp=false&q=false) <https://copycentralreaders.square.site/product/data-c104-human-contexts-ethics-of-data-edmundson-carson/7DPTZ2W7GZAOFFLJX67UN5V?cp=true&sa=true&sbp=false&q=false>. You will need to bring the course reader or your own printouts of the week's readings to each discussion section in order to fully participate. *Please note that not all required readings are in the course reader: in order to keep costs down, we included only the texts used for close reading activities in discussion section. Please consult the [course schedule \(https://bcourses.berkeley.edu/courses/1557692/pages/schedule\)](https://bcourses.berkeley.edu/courses/1557692/pages/schedule) to access PDFs of all course readings.*

You can find your section instructor's name and contact information on the [Discussion Sections and Office Hours page \(https://bcourses.berkeley.edu/courses/1557692/pages/discussion-sections-and-office-hours\)](https://bcourses.berkeley.edu/courses/1557692/pages/discussion-sections-and-office-hours), linked from the home menu.

Section Attendance and Participation (25%)

Section attendance is mandatory. The bulk of your section participation grade will come from consistent, active, and high-quality participation that demonstrates understanding of the prior week's lectures and reading materials, uses that course content to engage skillfully with questions and exercises from your section instructor, and contributes to other students' learning. Arriving late to section incurs a 20% penalty for that week, and may cause you to miss your warm up for the week. Failing to bring the course reader or printouts of the readings on a regular basis will also negatively impact your participation grade.

Please talk with your section instructor if you have questions about expectations for consistent, active, and high-quality participation. Each section has a unique set of people participating in it, and section instructors choose teaching strategies that are tuned for that section's learning.

Warm Ups (15%)

Preparation for section will be assessed by overall performance on written, closed-note "warm up" exercises that will take place during the first ten minutes of every section. Each warm up will ask you to respond to a specific prompt about one of the mandatory readings for the week or one of the main concepts from the previous week's lectures.

Warm ups will not be graded using the following scale.

Grade	Explanation
0	No submission; student writes nonsense or writes "I didn't do the reading."
5	Student demonstrates no evidence of having meaningfully engaged with the relevant course materials, but makes a sincere effort to discuss relevant concepts.
7	Student's response doesn't answer question but demonstrates good comprehension of other relevant material from the week <i>OR</i> the response answers the question but demonstrates poor comprehension of the material.
9	Student's response answers the question and demonstrates good comprehension of the material.
10	Student's response answers the question and demonstrates excellent comprehension of the material.

Students who do the readings carefully and pay attention in lecture should receive mostly 9's and 10's on their warm ups.

Warm ups are important to your learning, even though they follow a very simple format, because they prepare you for section and help you practice writing short answers about the course material before the exams. If you're not satisfied with your performance on your warm ups, please attend the tutor office hours.

You can skip up to three warm-ups or drop your three lowest warmup scores. The first warm-up is graded on completion. **Please note that warm-ups cannot be made up in case of an absence. Missed warm ups due to absences are covered by the three drops.**

Lectures

Lectures are a time for us to consider important issues all together as a class community. We encourage you to ask questions and share your perspectives. You will need to take notes while participating in the lecture. Lecture slides will be posted just before each lecture on the course schedule and in the media gallery.

Keeping up with lectures is essential to your ability to grasp course themes, participate in discussion section, and perform well on the exams. Lecture attendance is expected but not tracked.

Midterm 1 (10%)

A 90-minute in-person exam consisting of several short answer questions. It will be held on **Thursday, September 24th, beginning at 7pm.**

Midterm 2 (20%)

A 2-hour in-person exam consisting of two short answer questions and an analytical essay. It will be held on **Thursday, October 29th, beginning at 7pm.**

We will announce the exam locations closer to the date of the exams. Alternate exams and DSP accommodations will also be handled closer to the date of the exams.

Final exam (30%)

A 3-hour in-person final exam consisting of several short answer questions and an analytical essay. The exam will be held during the assigned final exam period for this class, **Tuesday, December 15th, beginning at 7pm.** The exam is comprehensive, but will emphasize content learned after the midterm.

Research Project (Alternative to Final Exam)

Students may choose to apply to submit a research project in place of the final exam. This project is a 2,000-word paper on a topic related to class content themes. It should apply key course concepts and readings to a focused research question while connecting your analysis to broader issues in the datafied world. The application process is selective, and acceptance of each project proposal is not guaranteed. Selection will depend on the quality of the application and the student's performance in the class to date.

Your project grade will be based on the quality of your research and writing, as well as your ability to integrate course concepts effectively. You are encouraged to consult your GSI when developing your topic.

Project Components and Deadlines:

1. Application: Topic proposal, including a motivation statement, that explains your choice and its connection to course concepts. Due Friday, October 9th, with final selections confirmed by Friday, October 16th.
2. Outline & Annotated Bibliography: A structured project outline along with a bibliography of 5–10 sources, each accompanied by 1–2 sentences explaining their content and relevance. Due Friday, October 30th.
3. Rough Draft: Submit a substantial draft of your paper followed by a peer review during the last week of classes. Due Monday, November 23rd.
4. Final Draft: Successful completion of this paper serves as an alternative to the final exam and excuses you from taking it. Due Tuesday, December 15th

Poll Everywhere

We use Poll Everywhere (PollEv) for in-class participation. [Sign up for Poll Everywhere \(https://rtl.berkeley.edu/services-programs/student-response-systems/students-getting-started-poll-everywhere\)](https://rtl.berkeley.edu/services-programs/student-response-systems/students-getting-started-poll-everywhere) before coming to the first class and [log in](https://pollev.com/login)  (https://pollev.com/login). PollEv requires internet access, so make sure you can access wifi, cell signal, etc. PollEv links for use in class: <https://pollev.com/ariel>  (https://pollev.com/ariel)

Grading

Your final course grade will be calculated as follows:

Final Grade Percentages

Category	Percentage of Grade
Section Attendance and Participation	25%
Section Warm Ups	15%
Midterm Exam 1	10%
Midterm Exam 2	20%
Final Exam	30%

Not all components are graded online and included in the online course grade book. Because of this, the online course grade book **will not display your overall course grade at any given time or your final grade**. It should simply be used to assess your performance on the components that are included within it. Your final letter grade will be mailed to you by the registrar's office or released on CalCentral approximately a week after the end of the course.

Grade Calibration

The course team calibrates across students and sections to provide an intersubjective basis for grading based on expert assessment. We calibrate each category of assignment, including section attendance and participation, and we calibrate again at the end of the course. We calibrate grades so that you can assume that variation in grades (across populations, such as from section to section, or from expectation to outcome) is due to variation in performance (achieving the goals of the assignment). This can be substantively hard for students to reckon with, because students don't yet have expert mastery of the material in order to assess variation in performance, and so sometimes make subjective attributions of variation in outcomes to variations in grading. Calibration is there to reduce the uncertainty in where to attribute variation in grades, and encourage you to focus on improving your performance.

The most effective way to improve your performance is to ask your section instructor for holistic feedback, based on **patterns** or **multiple aspects** of your performance. (Think of how you get expert feedback from an athletic coach or musical instructor.)

Some sections show stronger performance than others, and this is why different sections may have different mean grades. We don't have any limits on (say) the fraction of As we're prepared to give out. If the quality of your own work increases, or the performance of students overall increases, we will give higher grades.

Letter Grade Assignment

Letter grades will be assigned according to the schema below. Your final grade as listed on bCourses will be reviewed by your section instructor before final grades are submitted. Students just below a certain grade threshold (for example, 92.8) may have their final grade adjusted at the section instructor's discretion. Because every grade is reviewed in this way, we will not accept requests for grade changes in such circumstances.

Grade Distribution

Letter Grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
Range (%)	At instructor discretion	≥93 <99	≥90 <93	≥87 <90	≥83 <87	≥80 <83	≥77 <80	≥73 <77	≥70 <73	≥67 <70	≥63 <67	≥60 <63	<60

Course Policies

Grading Questions

If you have questions about warm up grading or exam grading (short answers), please attend the tutor office hours (see [office hours page \(https://bcourses.berkeley.edu/courses/1557692/pages/discussion-sections-and-office-hours\)](https://bcourses.berkeley.edu/courses/1557692/pages/discussion-sections-and-office-hours)). If you have questions about the grading of the essay components of the exams or about section participation grading, please contact your section instructor.

Questions about grades on assignments or exams should be asked at the time they are returned. We will not accept new questions or requests after 2 weeks.

Academic Integrity

The student community at UC Berkeley has adopted the following Honor Code: “As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.” The expectation is that you will adhere to this code.

Collaboration and Independence

Reviewing lecture and reading materials and studying for exams can be enjoyable and enriching things to do with fellow students. This is recommended. However, unless otherwise instructed, assignments are to be completed independently and materials submitted for them should be the result of one’s own independent work.

Cheating

A good lifetime strategy is always to act in such a way that no one would ever imagine that you would even consider cheating. Anyone caught cheating on an exam in this course will receive a failing grade in the course and will also be reported to the Center for Student Conduct. In order to guarantee that you are not suspected of cheating in an exam, please do not converse with others during the exams or copy and paste from materials outside this course. Anyone found to have submitted assignments completed by another person (student or non-student) will likewise receive a failing grade in the course and be reported to the Center for Student Conduct.

Plagiarism

To copy text or ideas from another source without appropriate reference is plagiarism and will result in a failing grade for the assignment and usually further disciplinary action. For additional information on plagiarism and how to avoid it see:

- [GSI Guide for Preventing Plagiarism \(https://gsi.berkeley.edu/gsi-guide-contents/academic-misconduct-intro/\)](https://gsi.berkeley.edu/gsi-guide-contents/academic-misconduct-intro/)

ChatGPT

We'll discuss ChatGPT and other publicly available text generating LLM's extensively in our course. Their uses are many and their social implications are complex. But we'll keep things relatively simple here:

Submitting work composed in whole or in part by ChatGPT counts as submitting work you did not do yourself and presenting as your own. It thus counts as plagiarism.

There are other uses of ChatGPT to aid in student learning that don't involve composing work and presenting it as your own. If you choose to use ChatGPT in one of these ways (for example, to help you brainstorm or outline ideas), you must disclose in writing that you used it and describe in several sentences *how* you used it. Just know that sometimes it gets things wrong, so use it at your own risk.

Also know that when ChatGPT generates language for topics specifically related to this course, it typically spits out bland, generic, and empty data/AI ethics boilerplate that is *not* what we teach. Because it's not what we teach, you can count on it getting a mediocre grade.

Turnitin

As a tool to promote academic integrity in this course, Turnitin will be used to check essays for originality. Turnitin takes assignments submitted through bCourses and compares them to a database of books, journal articles, websites, and other student papers. This creates an opportunity for you to improve your academic writing skills by ensuring that your sources have been properly cited and attributed. Turnitin can also help you gauge how much of your essay represents your own thinking and language. Please note that a low “Similarity Index” score does not necessarily indicate that the essay in question is without academic integrity issues. For more information on Turnitin see: [UC Berkeley Turnitin Page, Students Getting Started \(https://dls.berkeley.edu/services/academic-integrity/turnitin-students-getting-started\)](https://dls.berkeley.edu/services/academic-integrity/turnitin-students-getting-started)

Copyright notice about course materials

Course materials for this course, including all written materials and video, are protected by copyright. Prof. Edmundson, Prof. Carson, and Prof. Mursinna are the exclusive owners of copyright in those materials we create. You may not and may not allow others to reproduce or distribute these materials publicly whether or not a fee is charged without our express written consent. Reposting to third party sites or any other form of redistribution is prohibited.

Audio recording of any lecture or section material without the express permission of the instructors is prohibited, unless you are specifically entitled to record course materials through the Disabled Students Program.

Students with Disabilities

UC Berkeley is committed to creating a learning environment that meets the needs of its diverse student body including students with disabilities. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with the course instructors and your section instructor.

If you have a disability, or think you may have a disability, you can work with the Disabled Students' Program (DSP) to request an official accommodation. The Disabled Students' Program (DSP) is the campus office responsible for authorizing disability-related academic

accommodations, in cooperation with the students themselves and their instructors. You can find more information about DSP, including contact information and the application process here: [dsp.berkeley.edu \(http://dsp.berkeley.edu/\)](http://dsp.berkeley.edu). If you have already been approved for accommodations through DSP, the course instructors and your section instructor will have received your accommodation letter automatically. You should feel free to reach out to us to discuss implementation. There is no separate expectation that you reach out to us, however, in order to receive your accommodations.

Students who need academic accommodations or have questions about their accommodations should contact DSP, located at 100 Creekside Center. Students may call (510)-642-0518 (voice), 642-6376 (TTY), or e-mail [dsp@berkeley.edu \(mailto:dsp@berkeley.edu\)](mailto:dsp@berkeley.edu).

UC Berkeley is committed to providing robust educational experiences for all learners. With this goal in mind, we have activated the ALLY tool for this course. You will now be able to download content in a format that best fits your learning preference. PDF, HTML, EPUB, and MP3 are now available for most content items. For more information visit the alternative formats link or watch the video entitled, "[Ally in bCourses](https://youtu.be/dSkDtbQMKHA)  (<https://youtu.be/dSkDtbQMKHA>)



<https://youtu.be/dSkDtbQMKHA>

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Technical Support

If you are having technical difficulties, please alert your section instructor immediately. However, understand that neither the section instructors, nor the course instructors, can assist you with computer technical problems. You must call or email tech support and make sure you resolve any issues immediately. Be sure to document (save emails and transaction numbers) for all interactions with tech support. **Extensions and late submissions will not be accepted due to “technical difficulties.”**

For bCourses/Canvas technical support: click the Help button (in global navigation on the far left).

Course Summary:

Date	Details	Due
Thu Sep 24, 2026	 Midterm 1 (https://bcourses.berkeley.edu/courses/1557692/assignments/9121391)	due by 7pm
Thu Oct 29, 2026	 Midterm 2 (https://bcourses.berkeley.edu/courses/1557692/assignments/9121392)	due by 7pm
	 Academic Integrity Pledge (https://bcourses.berkeley.edu/courses/1557692/assignments/9120325)	
	 Key Course Policies Quiz (https://bcourses.berkeley.edu/courses/1557692/assignments/9120326)	
	 Orientation (https://bcourses.berkeley.edu/courses/1557692/assignments/9120327)	
	 Participation/Attendance Midterm (https://bcourses.berkeley.edu/courses/1557692/assignments/9120330)	
	 Warm Up #1 (9/1 - 9/3) (https://bcourses.berkeley.edu/courses/1557692/assignments/9120332)	
	 Warm Up #10 (11/3 - 11/5) (https://bcourses.berkeley.edu/courses/1557692/assignments/9120333)	

Date	Details	Due
	 Warm Up #11 (11/10 - 11/12) https://bcourses.berkeley.edu/courses/1557692/assignments/9120334	
	 Warm Up #12 (11/17 - 11/19) https://bcourses.berkeley.edu/courses/1557692/assignments/9120335	
	 Warm Up #13 (12/1 - 12/3) https://bcourses.berkeley.edu/courses/1557692/assignments/9120336	
	 Warm Up #2 (9/8 - 9/10) https://bcourses.berkeley.edu/courses/1557692/assignments/9120337	
	 Warm Up #3 (9/15 - 9/17) https://bcourses.berkeley.edu/courses/1557692/assignments/9120338	
	 Warm Up #4 (9/22 - 9/24) https://bcourses.berkeley.edu/courses/1557692/assignments/9120339	
	 Warm Up #5 (9/29 - 10/1) https://bcourses.berkeley.edu/courses/1557692/assignments/9124932	
	 Warm Up #6 (10/6 - 10/8) https://bcourses.berkeley.edu/courses/1557692/assignments/9120341	
	 Warm Up #7 (10/13 - 10/15) https://bcourses.berkeley.edu/courses/1557692/assignments/9120342	
	 Warm Up #8 (10/20 - 10/22) https://bcourses.berkeley.edu/courses/1557692/assignments/9120343	
	 Warm Up #9 (10/27 - 10/29) https://bcourses.berkeley.edu/courses/1557692/assignments/9120344	